PSEUDO-COHERENT DIRECT IMAGE OF A TENSOR PRODUCT WITH A BOUNDED ABOVE FLAT FINITELY PRESENTED COMPLEX OF PROPER SUPPORT COMMUTES WITH BASE CHANGE

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Allowed standard prerequisites (editorial).

- The formation of the derived pushforward of a tensor product commutes with arbitrary base change under the stated hypotheses, and over the perfect case the result object is perfect. invoked at proof lines 7689-7692 and again at proof lines 7736-7737. Reduces the lemma to m-pseudo-coherence and provides the perfect source object.
- The total direct image of a quasi-compact quasi-separated morphism has a finite cohomological dimension bound, recorded with the quasi-coherence lemma. invoked at proof lines 7697-7701. Supplies the shift amount N entering every truncation degree.
- Approximation by perfect complexes: for suitable parameters there is a perfect complex P with a map P -> E isomorphic in cohomology in high degrees, by the approximation theorem. invoked at proof lines 7704-7706. Provides the perfect approximation P of E.

Source reference. Original lemma 0A1D: sources/stacks-perfect/pinned-original-source.tex, source lines 6880–6880. Cited at proof line 7689 to dispose of the base change half; the pseudo-coherence half is inside the counted span.

Source reference. Original lemma 0A1H: sources/stacks-perfect/pinned-original-source.tex, source lines 7614–7614. Cited at proof lines 7691-7692 for the strategy and at proof lines 7736-7737 to conclude the truncated source is perfect; the approximation argument is inside the counted span.

Source reference. Original lemma 08D5: sources/stacks-perfect/pinned-original-source.tex, source lines 661–661. Cited at proof lines 7700-7701 for the cohomological dimension bound N; the degree chases using N are inside the counted span.

Source reference. Original theorem 08ES: sources/stacks-perfect/pinned-original-source.tex, source lines 4031–4031. Cited at proof lines 7705-7706 to obtain the

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perfect approximation P -> E; the degree chases using P are inside the counted span.

Lemma 0.2. *Let $f : X \to S$ be a morphism of finite presentation. Let $E \in D(\mathcal{O}_X)$ be a pseudo-coherent object. Let $\mathcal{G}^\bullet$ be a bounded above complex of finitely presented $\mathcal{O}_X$-modules, flat over S, with support proper over S. Then*

$$K = Rf_*(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{G}^\bullet)$$

is a pseudo-coherent object of $D(\mathcal{O}_S)$ and its formation commutes with arbitrary base change.

Proof. The statement on base change is Lemma 0A1D. Thus it suffices to show that K is a pseudo-coherent object. This will follow from Lemma 0A1H by approximation by perfect complexes. We encourage the reader to skip the rest of the proof.

The question is local on S, hence we may assume S is affine. Then X is quasi-compact and quasi-separated. Moreover, there exists an integer N such that total direct image $Rf_* : D_{QCoh}(\mathcal{O}_X) \to D_{QCoh}(\mathcal{O}_S)$ has cohomological dimension N as explained in Lemma 08D5. Choose an integer b such that $\mathcal{G}^i = 0$ for $i > b$. It suffices to show that K is m-pseudo-coherent for every m. Choose an approximation $P \to E$ by a perfect complex P of $(X, E, m-N-1-b)$. This is possible by Theorem 08ES. Choose a distinguished triangle

$$P \to E \to C \to P[1]$$

in $D_{QCoh}(\mathcal{O}_X)$. The cohomology sheaves of C are zero in degrees $\geq m-N-1-b$. Hence the cohomology sheaves of $C \otimes^{\mathbf{L}} \mathcal{G}^\bullet$ are zero in degrees $\geq m-N-1$. Thus the cohomology sheaves of $Rf_*(C \otimes^{\mathbf{L}} \mathcal{G}^\bullet)$ are zero in degrees $\geq m-1$. Hence

$$Rf_*(P \otimes^{\mathbf{L}} \mathcal{G}^\bullet) \to Rf_*(E \otimes^{\mathbf{L}} \mathcal{G}^\bullet)$$

is an isomorphism on cohomology sheaves in degrees $\geq m$. Next, suppose that $H^i(P) = 0$ for $i > a$. Then $P \otimes^{\mathbf{L}} \sigma_{\geq m-N-1-a}\mathcal{G}^\bullet \longrightarrow P \otimes^{\mathbf{L}} \mathcal{G}^\bullet$ is an isomorphism on cohomology sheaves in degrees $\geq m-N-1$. Thus again we find that

$$Rf_*(P \otimes^{\mathbf{L}} \sigma_{\geq m-N-1-a}\mathcal{G}^\bullet) \to Rf_*(P \otimes^{\mathbf{L}} \mathcal{G}^\bullet)$$

is an isomorphism on cohomology sheaves in degrees $\geq m$. By Lemma 0A1H the source is a perfect complex. We conclude that K is m-pseudo-coherent as desired. □